

Ryan Cho

Software Engineer | Data Engineering & Analytics Platforms

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PROFESSIONAL SUMMARY

Software engineer with 7+ years building data pipelines, transformation layers, and operational analytics applications in financial services and SaaS. Strong in Python, Java, and Spark for distributed processing, with hands-on experience integrating source feeds, validating datasets, and delivering TypeScript and JavaScript tooling for cross-functional teams. Known for debugging build failures and translating schema and lineage questions into reliable production systems.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Owned **end-to-end ingestion pipelines** moving trading, ledger, and reference feeds from upstream sources into a centralized analytics store supporting 40+ daily operational reports.
- Built **PySpark transformation jobs** to normalize, deduplicate, and derive portfolio risk datasets, reducing manual reconciliation cycles for downstream analytics consumers.
- Designed **Python and Java orchestration services** for scheduled batch loads that enforce schema contracts and expose build health metrics to engineering and operations stakeholders.
- Partnered with product and compliance teams to **debug failed builds**, trace lineage gaps, and deliver accurate responses to platform data footprint requests under regulatory deadlines.
- Implemented **Pandas validation layers** for ad hoc operational inquiries, enabling analysts to investigate exposure and cash-position questions without direct warehouse access.
- Developed **TypeScript dashboards and internal APIs** exposing curated datasets and supporting iterative debugging of core platform logic alongside feature teams.
- Introduced **monitoring and alerting** around pipeline SLA breaches, shortening mean time to detect ingestion regressions across nightly batch windows.
- Mentored engineers on **distributed data patterns**, Spark partitioning strategies, and safe deployment practices for high-volume financial datasets.
- Refactored legacy ETL scripts into **modular Spark stages**, improving maintainability and enabling parallel execution over multi-terabyte historical archives.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Supported **data feed transfers** from core banking and market-data sources into internal risk and reporting environments used by global operations teams.
- Authored **Java batch processors** and **Python utilities** transforming trade and position extracts into governed datasets aligned with firm-wide schema standards.
- Monitored nightly **build pipelines**, triaged transformation failures, and coordinated fixes with upstream system owners to restore SLA-aligned delivery.
- Collaborated with analysts to design **operational reporting applications** answering exposure, liquidity, and counterparty questions from consolidated warehouse tables.
- Responded to **information requests** on data footprint, retention, and field definitions, documenting lineage for audit and regulatory reviews.
- Worked with cross-functional squads to **trace application behavior** through SQL, Java services, and messaging layers during incident investigations.
- Applied **algorithms and data structures** to optimize join-heavy aggregations over large position histories, improving runtime on critical month-end batch jobs.

- Built **JavaScript tooling** for lightweight data quality checks and prototype views supporting technology modernization initiatives.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **Python ingestion scripts** pulling restaurant compliance data from CSV exports and APIs into a normalized store for reporting workflows.
- Developed **JavaScript and React components** displaying operational metrics to venue managers through backend REST endpoints.
- Assisted in **debugging ETL jobs** syncing nightly transaction batches, documenting failure patterns for the engineering team.
- Implemented **data validation checks** on incoming feed records, flagging schema mismatches before they reached production dashboards.
- Collaborated with product stakeholders to clarify **operational reporting requirements** and iterate on dashboard filters and export formats.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Created **JavaScript admin views** for a nonprofit donor database, improving data capture consistency across fundraising campaigns.
- Integrated third-party **data imports** into the organization's CMS, transforming spreadsheet uploads into structured records for outreach teams.
- Maintained **Python utilities** for cleaning contact lists and deduplicating entries prior to mail-merge operations.
- Supported stakeholders with **ad hoc data extracts** and operational reports drawn from the foundation's internal database.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: TypeScript, JavaScript, React, HTML5, CSS3

Backend: Python, Java, Node.js, PySpark, Apache Spark, REST APIs

Databases: PostgreSQL, MySQL, SQL Server, SQL, data warehousing

Cloud & DevOps: AWS, Docker, CI/CD, Apache Airflow, Linux

Monitoring & Tools: Pandas, pipeline observability, logging and alerting, Git, Jira

Workflow & PM: Agile/Scrum, cross-functional collaboration, stakeholder communication, technical documentation